ClearFund

Transparent Fund Disbursement and Misuse Detection System

Abstract

Public funds such as government subsidies, disaster relief, and welfare payments are currently disbursed as cash or standard bank transfers, offering no mechanism to control, verify, or trace how the money is ultimately spent, which is a major contributor to fund leakage and misuse. ClearFund addresses this by issuing public funds as digital credits with built-in spending rules enforced by the system itself. Beneficiaries can only spend these credits at verified vendors within approved categories, and every transaction is recorded in a tamper-evident ledger, where each entry is cryptographically linked to the one before it, making any alteration to past records immediately detectable.

The system flags unusual transaction patterns, such as repeated transfers between the same accounts in a short period, to help identify possible misuse. A simple public dashboard shows how funds are being utilized overall, by category, without revealing any individual beneficiary's personal transaction details. By combining rule-based spending restrictions, automatic validation, and a tamper-evident transaction record, ClearFund offers a practical way to improve transparency and accountability in public fund disbursement.

Users

Role	Description
Government (Issuer / Admin)	Issues fund credits, defines spending rules per fund type, monitors disbursement and reviews flagged transactions
Beneficiary (Citizen)	Receives fund credits in a digital wallet and spends them at approved vendors
Vendor	Registered to accept restricted-category fund credits from beneficiaries

Key Features

- Spending restrictions on issued funds, enforced by the system's rules engine
- Automatic transaction validation with no manual approval step required
- Tamper-evident transaction ledger using cryptographically linked records
- Rule-based flagging of unusual transaction patterns
- Vendor registration for accepting restricted-category funds
- Public dashboard showing aggregate fund utilization by category
- Government dashboard for fund issuance and reviewing flagged transactions

Technology Stack

- Backend: Python, FastAPI
- Ledger: Python — hash-chained transaction records (SHA-256) stored in PostgreSQL
- Rules & Flagging Logic: Python
- Database: PostgreSQL
- Frontend: React / Next.js